PLACEMENT PORTAL APPLICATION

MAD 1 PROJECT

1. Student Details

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2. Project Details

Project Title: Placement Portal Application

Problem Statement:

Many institutes struggle to efficiently manage campus recruitment due to dependency on spreadsheets, emails, and manual processes. This leads to difficulties in tracking student applications, managing company approvals, avoiding duplicate registrations, and maintaining accurate placement records. A centralized web-based Placement Portal is needed to streamline these activities for admins, companies, and students.

Approach:

To solve this issue, a web-based Placement Portal was developed using the Flask framework with a modular design in VS Code. The system supports three distinct user roles: students, companies, and administrators. Students can sign up, build their profiles, explore job openings, apply for positions, and track their application progress. Companies can create profiles, post job drives with detailed job information, and review student applications. The administrator oversees the platform by approving registrations, managing users, and monitoring placement activities to ensure smooth operation.

3. AI/LLM Declaration

In developing the Placement Portal, ChatGPT (GPT-5) was utilized to provide guidance in creating SQL Alchemy model definitions and improving the consistency of variable naming across the project. To strengthen understanding and implementation skills, various YouTube tutorials were also referenced, offering practical insights into the technologies used.

The AI contribution was limited to around 20%, focusing solely on code suggestions and formatting support.

All core development tasks, including writing the main logic, debugging, testing, and integrating the system, were performed manually to ensure complete control and reliability of the application.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
Flask-SQL Alchemy	Database integration for Flask
SQL Alchemy	Object Relational Mapper (ORM) for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap	Frontend styling and responsive design
Flask-Login	User authentication and session management
Python-dotenv	Used to manage environment variables

5. Database Schema / ER Diagram

Tables

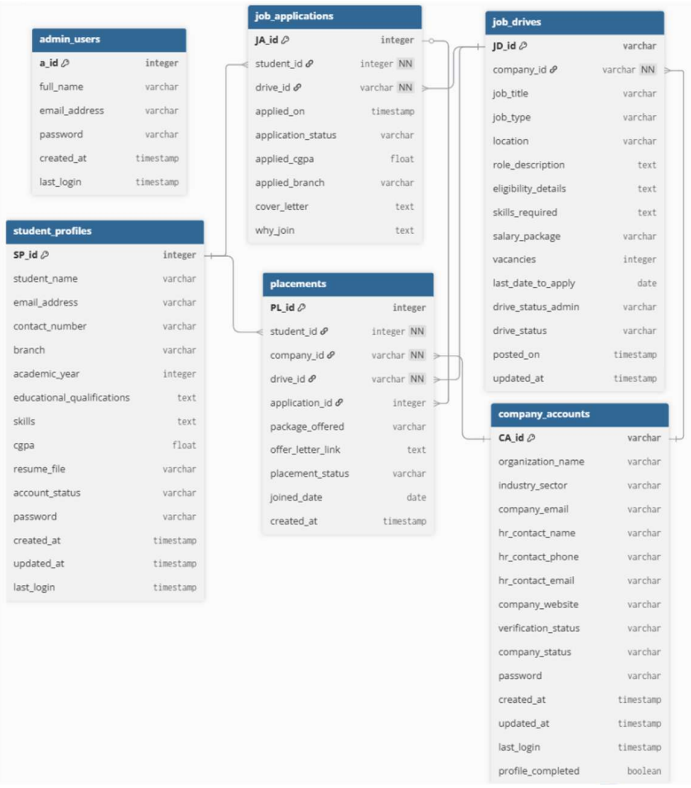
AdminUser-Stores administrator login and management details.

StudentProfile- Stores student personal, academic, and skill information.

CompanyAccount-Stores company details and HR contact information.

JobDrive-Stores job drives posted by companies.

JobApplication-Stores applications submitted by students for job drives.



Placement-Stores final placement details of selected students.

Relationship Overview

- **CompanyAccount → JobDrive** : One-to-Many
- **StudentProfile → JobApplication** : One-to-Many
- **JobDrive → JobApplication** : One-to-Many
- **StudentProfile → Placement:** One-to-Many
- **JobApplication → Placement:** One-to-One

6. Web Routes and Endpoints

Authentication

- **/** – Home page
- **/login** – Allows users (Admin, Student, Company) to log in
- **/register** – Allows new students or companies to register
- **/logout** – Logs the user out of the system

Student

- **/student/dashboard** – Displays student dashboard
- **/student/update_profile** – View or update student profile
- **/student/show_drives** – Displays available job drives
- **/student/jobs/<JD_id>/apply** – Apply for a job drive
- **/student/applications_status** – View application status
- **/student/placement_history** – View placement history

Company

- **/company/dashboard** – Displays company dashboard
- **/company/update_profile** – Complete or update company profile
- **/company/create_jobs** – Post a new job drive
- **/company/drives** – View company job drives
- **/company/drives/<drive_id>/applications** – View applications for a specific drive
- **/company/application/<app_id>/<status>** – Update application status (Shortlisted, Interview, Rejected, Selected)
- **/company/drives/close/<JD_id>** – Close a job drive
- **/company/drives/activate/<JD_id>** – Activate a job drive

Admin

- **/admin/dashboard** – Admin overview dashboard
- **/admin/students/manage_students** – Manage student accounts
- **/admin/companies/manage_companies** – Manage company accounts

- **/admin/drives/manage_drives** – Manage job drives
- **/admin/applications/manage_applications** – View and manage all job applications

Additional System Routes

- **/uploads/resumes/<filename>** – View student resume files
- **/profile/student/<student_id>** – View student profile (Admin / Company access)

7. Architecture Overview

The Placement Portal is built using **Flask with a structured MVC-like design**. The application separates database models, application routes, templates, and static resources to keep the system organized and maintainable.

Project Components

app.py
Initializes the Flask application and connects the database.

models.py
Defines the core database tables using SQLAlchemy: AdminUser, StudentProfile, CompanyAccount, JobDrive, JobApplication, and Placement.

routes.py
Handles all system functionalities such as login, registration, dashboards, job drives, job applications, and placement management.

templates/
Contains HTML pages rendered using Jinja2 for admin, student, and company interfaces.

static/
Includes CSS, JavaScript, Bootstrap files, and uploaded resumes used for the frontend design.

Core Features

- Secure authentication and registration system
- Role-based system for **Admin, Company, and Student**
- Students can create profiles and apply for job drives
- Companies can post and manage placement drives
- Admin can manage students, companies, drives, and applications
- Application progress tracking from **Applied to Placed**

Additional Capabilities

- Admin controls for enabling/disabling students and companies
- Automatic closing of expired job drives
- Placement records for selected students
- Search options for drives, students, companies, and applications

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/132KMTtXxlZv07xjVRapFUYPkNDcVymo7/view?usp=sharing